

Faculty Management System

Group Members:

Ariba Fatima 133690
Areej Fatima 133671
Maha Irfan 133667

Submitted to:

Ms. Sehrish Khan

Course:

CC-215L-Database

```
mysql> show databases;
```

Database
aribadb
empl
employee
information_schema
mysql
performance_schema
sys

```
7 rows in set (0.10 sec)
```

```
mysql> CREATE DATABASE FMS;
```

```
Query OK, 1 row affected (0.08 sec)
```

```
mysql> show databases;
```

Database
aribadb
empl
employee
fms
information_schema
mysql
performance_schema
sys

```
8 rows in set (0.00 sec)
```

Admin:

```
mysql> CREATE TABLE Admin(Admin_ID INT(5) PRIMARY KEY, Admin_Name VARCHAR(50) NOT NULL, A_phno VARCHAR(30) UNIQUE NOT NULL, A_Password VARCHAR(30) NOT NULL, A_Email VARCHAR(40) UNIQUE);
ERROR 1050 (42S01): Table 'admin' already exists
mysql> show tables;
+-----+
| Tables_in_fms |
+-----+
| admin          |
+-----+
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO Admin Values(1, 'Ali', '12345', 'alihere', 'ali123', 'ali@gmail.com');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> select * from Admin;
```

Admin_ID	Admin_Name	A_phno	A_Username	A_Password	A_Email
1	Ali	12345	alihere	ali123	ali@gmail.com

```
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO Admin Values(2, 'Asif', '56442', 'iasif', '09asif', 'asif@gmail.com');
Query OK, 1 row affected (0.04 sec)
```

```
mysql> select * from Admin;
```

Admin_ID	Admin_Name	A_phno	A_Username	A_Password	A_Email
1	Ali	12345	alihere	ali123	ali@gmail.com
2	Asif	56442	iasif	09asif	asif@gmail.com

```
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Admin Values(3, 'Zain', '09876', 'mezain', 'zain223', 'zain@gmail.com');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from Admin;
```

Admin_ID	Admin_Name	A_phno	A_Username	A_Password	A_Email
1	Ali	12345	alihere	ali123	ali@gmail.com
2	Asif	56442	iasif	09asif	asif@gmail.com
3	Zain	09876	mezain	zain223	zain@gmail.com

```
3 rows in set (0.00 sec)
```

Department:

```
mysql> CREATE TABLE Department(Deptt_ID INT(8) PRIMARY KEY, Deptt_Name VARCHAR(50) NOT NULL , Deptt_Location VARCHAR(50) , HOD VARCHAR(40) NOT NULL, Admin_ID INT, FOREIGN KEY (Admin_ID) REFERENCES Admin(Admin_ID) ON DELETE SET NULL ON UPDATE CASCADE);
Query OK, 0 rows affected, 1 warning (0.09 sec)
```

```
mysql> show tables;
```

```
+-----+
| Tables_in_fms |
+-----+
| admin          |
| department     |
+-----+
```

```
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Department(Deptt_ID, Deptt_Name, Deptt_Location, HOD, Admin_ID) Values (101, 'Information Technology', 'Near Staff Room', 'Prof. Ali Waqas', 1);
Query OK, 1 row affected (0.04 sec)
```

```
mysql> select * from Department;
```

```
+-----+-----+-----+-----+-----+
| Deptt_ID | Deptt_Name          | Deptt_Location | HOD          | Admin_ID |
+-----+-----+-----+-----+-----+
| 101      | Information Technology | Near Staff Room | Prof. Ali Waqas | 1        |
+-----+-----+-----+-----+-----+
```

```
1 row in set (0.04 sec)
```

```
mysql> INSERT INTO Department(Deptt_ID, Deptt_Name, Deptt_Location, HOD, Admin_ID) Values (102, 'Mathematics', 'Bloack B', 'Prof. Sara a Azam', 2);
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from Department;
```

```
+-----+-----+-----+-----+-----+
| Deptt_ID | Deptt_Name          | Deptt_Location | HOD          | Admin_ID |
+-----+-----+-----+-----+-----+
| 101      | Information Technology | Near Staff Room | Prof. Ali Waqas | 1        |
| 102      | Mathematics          | Bloack B       | Prof. Sara Azam | 2        |
+-----+-----+-----+-----+-----+
```

```
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Department(Deptt_ID, Deptt_Name, Deptt_Location, HOD, Admin_ID) Values (103, 'Zoology', 'Block C', 'Prof. Nadia Khan', 3);
Query OK, 1 row affected (0.04 sec)
```

```
mysql> select * from Department;
```

```
+-----+-----+-----+-----+-----+
| Deptt_ID | Deptt_Name          | Deptt_Location | HOD          | Admin_ID |
+-----+-----+-----+-----+-----+
| 101      | Information Technology | Near Staff Room | Prof. Ali Waqas | 1        |
| 102      | Mathematics          | Bloack B       | Prof. Sara Azam | 2        |
| 103      | Zoology              | Block C        | Prof. Nadia Khan | 3        |
+-----+-----+-----+-----+-----+
```

```
3 rows in set (0.00 sec)
```


Course:

```
mysql> CREATE TABLE Course(C_ID VARCHAR(10) UNIQUE NOT NULL,C_Name VARCHAR(50) UNIQUE NOT NULL,Credit_Hours VARCHAR(10) NOT NULL,FAC_ID INT,FOREIGN KEY (FAC_ID) REFERENCES Faculty (FAC_ID) ON DELETE SET NULL ON UPDATE CASCADE, Deptt_ID INT, FOREIGN KEY (Deptt_ID) REFERENCES Department(Deptt_ID) ON DELETE SET NULL ON UPDATE CASCADE);
Query OK, 0 rows affected (0.13 sec)

mysql> show tables;
+-----+
| Tables_in_fms |
+-----+
| admin          |
| course         |
| department     |
| faculty        |
+-----+
4 rows in set (0.00 sec)

mysql> INSERT INTO Course(C_ID, C_Name, Credit_Hours, FAC_ID, Deptt_ID) Values ('111', 'Software Engineering', '3', 201, 101);
Query OK, 1 row affected (0.06 sec)

mysql> select * from Course;
+-----+-----+-----+-----+-----+
| C_ID | C_Name                | Credit_Hours | FAC_ID | Deptt_ID |
+-----+-----+-----+-----+-----+
| 111  | Software Engineering  | 3            | 201    | 101      |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO Course(C_ID, C_Name, Credit_Hours, FAC_ID, Deptt_ID) Values ('112', 'Digital Logic and Design', '3', 201, 101);
Query OK, 1 row affected (0.04 sec)

mysql> select * from Course;
+-----+-----+-----+-----+-----+
| C_ID | C_Name                | Credit_Hours | FAC_ID | Deptt_ID |
+-----+-----+-----+-----+-----+
| 111  | Software Engineering  | 3            | 201    | 101      |
| 112  | Digital Logic and Design | 3            | 201    | 101      |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> INSERT INTO Course(C_ID, C_Name, Credit_Hours, FAC_ID, Deptt_ID) Values ('113', 'Calculus and Analytical Geometry', '3', 203, 102);
Query OK, 1 row affected (0.04 sec)

mysql> select * from Course;
```

```
mysql> select * from Course;
+-----+-----+-----+-----+-----+
| C_ID | C_Name                | Credit_Hours | FAC_ID | Deptt_ID |
+-----+-----+-----+-----+-----+
| 111  | Software Engineering  | 3            | 201    | 101      |
| 112  | Digital Logic and Design | 3            | 201    | 101      |
| 113  | Calculus and Analytical Geometry | 3            | 203    | 102      |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Attendance Record:

```
mysql> CREATE TABLE Att_rec(Attd_ID INT(10) PRIMARY KEY, Attd_Date VARCHAR(50) NOT NULL, Status VARCHAR(15) CHECK (Status IN ('Present','Absent','Leave')),
FAC_ID INT, FOREIGN KEY (FAC_ID) REFERENCES Faculty (FAC_ID) ON DELETE SET NULL ON UPDATE CASCADE, Admin_ID INT, FOREIGN KEY (Admin_ID) REFERENCES Admin (Ad
min_ID) ON DELETE SET NULL ON UPDATE CASCADE);
Query OK, 0 rows affected, 1 warning (0.10 sec)
```

```
mysql> show tables;
```

Tables_in_fms
admin
att_rec
course
department
faculty

```
5 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Att_rec(Attd_ID, Attd_Date, Status, FAC_ID, Admin_ID) Values (1001, '27/10/2025', 'Present', 201, 2);
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from Att_rec;
```

Attd_ID	Attd_Date	Status	FAC_ID	Admin_ID
1001	27/10/2025	Present	201	2

```
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO Att_rec(Attd_ID, Attd_Date, Status, FAC_ID, Admin_ID) Values (1002, '28/10/2025', 'Leave', 201, 2);
Query OK, 1 row affected (0.04 sec)
```

```
mysql> select * from Att_rec;
```

Attd_ID	Attd_Date	Status	FAC_ID	Admin_ID
1001	27/10/2025	Present	201	2
1002	28/10/2025	Leave	201	2

```
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Att_rec(Attd_ID, Attd_Date, Status, FAC_ID, Admin_ID) Values (1003, '28/10/2025', 'Present', 202, 3);
Query OK, 1 row affected (0.04 sec)
```

```
mysql> select * from Att_rec;
```

Attd_ID	Attd_Date	Status	FAC_ID	Admin_ID
1001	27/10/2025	Present	201	2
1002	28/10/2025	Leave	201	2
1003	28/10/2025	Present	202	3

```
3 rows in set (0.00 sec)
```

Leave Request:

```
mysql> CREATE TABLE Leave_Request(Leave_ID INT(10) PRIMARY KEY, Status VARCHAR(15) CHECK (Status IN ('Pending','Approved','Rejected'))
), Reason VARCHAR(1000), Start_Date VARCHAR(20) NOT NULL, End_Date VARCHAR(20) NOT NULL , FAC_ID INT, FOREIGN KEY (FAC_ID) REFERENCES
Faculty (FAC_ID) ON DELETE SET NULL ON UPDATE CASCADE, Admin_ID INT, FOREIGN KEY (Admin_ID) REFERENCES Admin (Admin_ID) ON DELETE SE
T NULL ON UPDATE CASCADE);
Query OK, 0 rows affected, 1 warning (0.08 sec)
```

```
mysql> show tables;
+-----+
| Tables_in_fms |
+-----+
| admin          |
| att_rec        |
| course         |
| department     |
| faculty        |
| leave_request  |
+-----+
6 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Leave_request(Leave_ID, Status, Reason, Start_Date,End_Date, FAC_ID, Admin_ID) Values (2001, 'Pending','illness',
'27/10/2025','30/10/2025', 203, 3);
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from Leave_request;
+-----+-----+-----+-----+-----+-----+-----+
| Leave_ID | Status | Reason | Start_Date | End_Date | FAC_ID | Admin_ID |
+-----+-----+-----+-----+-----+-----+-----+
| 2001 | Pending | illness | 27/10/2025 | 30/10/2025 | 203 | 3 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO Leave_request(Leave_ID, Status, Reason, Start_Date,End_Date, FAC_ID, Admin_ID) Values (2002, 'Approved','illness',
'26/10/2025','28/10/2025', 201, 3);
Query OK, 1 row affected (0.04 sec)
```

```
mysql> select * from Leave_request;
+-----+-----+-----+-----+-----+-----+-----+
| Leave_ID | Status | Reason | Start_Date | End_Date | FAC_ID | Admin_ID |
+-----+-----+-----+-----+-----+-----+-----+
| 2001 | Pending | illness | 27/10/2025 | 30/10/2025 | 203 | 3 |
| 2002 | Approved | illness | 26/10/2025 | 28/10/2025 | 201 | 3 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Leave_request(Leave_ID, Status, Reason, Start_Date,End_Date, FAC_ID, Admin_ID) Values (2003, 'Rejected','Tour', '2
9/10/2025','1/11/2025', 202, 3);
Query OK, 1 row affected (0.04 sec)
```

```
mysql> select * from Leave_request;
+-----+-----+-----+-----+-----+-----+-----+
| Leave_ID | Status | Reason | Start_Date | End_Date | FAC_ID | Admin_ID |
+-----+-----+-----+-----+-----+-----+-----+
| 2001 | Pending | illness | 27/10/2025 | 30/10/2025 | 203 | 3 |
| 2002 | Approved | illness | 26/10/2025 | 28/10/2025 | 201 | 3 |
| 2003 | Rejected | Tour | 29/10/2025 | 1/11/2025 | 202 | 3 |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Salary:

```
mysql> CREATE TABLE Salary(Payroll_ID INT(10) PRIMARY KEY, Basic_Salary INT(10) NOT NULL, Allowances INT(10) NOT NULL, Deduction INT(10) NOT NULL, Payment_Date VARCHAR(20) NOT NULL, FAC_ID INT, FOREIGN KEY (FAC_ID) REFERENCES Faculty (FAC_ID) ON DELETE SET NULL ON UPDATE CASCADE);
Query OK, 0 rows affected, 4 warnings (0.07 sec)
```

```
mysql> show tables;
```

Tables_in_fms
admin
att_rec
course
department
faculty
leave_request
salary

```
7 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Salary (Payroll_ID, Basic_Salary, Allowances, Deduction, Payment_Date, FAC_ID) VALUES (3001, 50000, 10000, 5000, '2025-10-27', 201);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from Salary;
```

Payroll_ID	Basic_Salary	Allowances	Deduction	Payment_Date	FAC_ID
3001	50000	10000	5000	2025-10-27	201

```
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO Salary (Payroll_ID, Basic_Salary, Allowances, Deduction, Payment_Date, FAC_ID) VALUES (3002, 45000, 10000, 5000, '2025-10-27', 202);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from Salary;
```

Payroll_ID	Basic_Salary	Allowances	Deduction	Payment_Date	FAC_ID
3001	50000	10000	5000	2025-10-27	201
3002	45000	10000	5000	2025-10-27	202

```
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Salary (Payroll_ID, Basic_Salary, Allowances, Deduction, Payment_Date, FAC_ID) VALUES (3003, 40000, 10000, 5000, '2025-10-27', 203);
Query OK, 1 row affected (0.04 sec)
```

```
mysql> select * from Salary;
```

Payroll_ID	Basic_Salary	Allowances	Deduction	Payment_Date	FAC_ID
3001	50000	10000	5000	2025-10-27	201
3002	45000	10000	5000	2025-10-27	202
3003	40000	10000	5000	2025-10-27	203

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT Payroll_ID, Basic_Salary, Allowances, Deduction, (Basic_Salary + Allowances - Deduction) AS Net_Salary FROM Salary;
```

Payroll_ID	Basic_Salary	Allowances	Deduction	Net_Salary
3001	50000	10000	5000	55000
3002	45000	10000	5000	50000
3003	40000	10000	5000	45000

```
3 rows in set (0.01 sec)
```